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Optical performance characterization of light-logging actigraphy dosimeters

Luke L. A. Price,* Andrey Lyachev, and Marina Khazova

Public Health England, Centre for Radiation, Chemical and Environmental Hazards, Didcot, Oxfordshire OX11 0RQ, UK
**Corresponding author: Luke. Price@phe.gov.uk*

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There are several wearable products specially developed or marketed for studying sleep, circadian rhythms, and light levels. However, new recommendations relating to human physiological responses to light have changed what measurements researchers may demand. The performances of 11 light-logging dosimeters from eight manufacturers were compared. The directional and spectral sensitivities, linearity, dynamic range, and resolution were tested for seven models, and compared along with other published data. The sample mainly comprised light-logging actigraphy dosimeters wearable as badges, in accordance with measurement protocols for larger-scale field studies. A proposed standard for optical performance assessments is set out.

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1. INTRODUCTION

Many products are available that log visible light conditions, which could be used in large-scale field studies into the effects of light. The majority are designed to measure illuminance, a quantity of light based on the visual system. However, there are other "non-visual" physiological responses to visible light, with different measurement requirements, particularly for spectral sensitivity. Unfortunately, spectral measurements are not usually practical in field studies requiring wearable devices, as current spectroradiometers are bulky and usually require separate power, computer control, and data storage.

The purpose of the study is to systematically characterize the optical performance of a range of actigraphy dosimeters in current use in field studies concerning the new recommendations for measurements related to the non-visual effects of light on health. There are no previous data in the literature for several of the dosimeters and this is the first study of more than three models.

Most, if not all, dosimeters designed to study physiological responses to light also include activity sensors. Many of these actigraphs incorporating light sensors were designed to mimic the spectral sensitivity of the human visual system. This is perhaps because the visual system has been almost exclusively used to characterize other wearable light sensors in the past. There are also designs utilizing historically favored spectral sensitivities for non-visual responses in people. It is reasonable to presume that sleep actigraphs incorporating light sensors are intended for the study of non-visual physiological responses, rather than visual responses to light. This may apply to a lesser extent to sports research applications of actigraphs. It is important to appreciate that what a device actually does and the range of different ways it might be used are more important selection criteria than what it was originally intended to do.

In 2014, the First International Workshop on Circadian Neurophysiological Photoreception, 2013 (IWCNP), a panel of representatives from leading research groups on human non-visual action spectra, published consensus recommendations regarding the measurement of light with respect to non-visual responses [1]. The proposed system entails measuring light according to the five different "α-opic" spectral sensitivities of the five photoreceptor types in the human eye, all of which provide light information into the non-visual system. It was intended to replace former contradictory approaches, particularly with respect to the various spectral sensitivities and metrics used to report light stimuli.

The IWCNP recommendations were the subject of a supportive statement and a technical publication by the International Commission on Illumination (CIE) [2,3]. CIE is expected to adopt the five sensitivity curves in a measurement standard for studying human non-visual responses to light. Accordingly, the current IWCNP and CIE recommendations are followed here, and the results apply to circadian and non-visual research in general. No dosimeters have yet been based on the recent recommendations.

These recommendations capture all potential non-visual stimulus information. Other non-visual action spectra proposed before IWCNP may turn out to be valid. Despite this,

IWCNP recommended establishing the validity of other approaches through first carrying out systematic studies using the five α-opic sensitivity curves.

In 1987, CIE published Methods of Characterizing Illuminance Meters and Luminance Meters [4], intended for evaluating their performance (the "CIE report"). These methods have more recently been applied [5–9] to characterize dosimeters. An attempt to generalize and adapt the methods for circadian and non-visual dosimetry, and to a limited extent optical broadband metrology in general, is made here.

While the use of the α-opic curves and the proposed characterization methods are independently justified, they are closely linked in this application as essential links in the dosimetry chain. The dosimetry chain [10] describes a set of considerations governing reproducibility in non-visual dosimetry studies. Reproducibility demands traceability, and ideally consensus, for non-visual metrology (e.g., Lucas *et al.* [1]) and device characterization, as well as personal light exposure measurement protocols and exposure time series metrics.

The current study is restricted to characterization methods based on the α-opic quantities endorsed by CIE and comparing these to existing CIE characterization methods for illuminance meters. Detailed comments on personal light exposure measurement protocols, metrics derived from exposure time series [11–13], and how to analyze effects on health and well-being are excluded from the scope.

The study results make it possible for researchers and practitioners to better understand the contribution of optical performance to measurement errors from devices in current use. Further sources of measurement errors, which should not be overlooked when designing and interpreting field studies, include subject compliance levels, light-based eye and head movement, and the general influence of device measurement position.

Polarization response, aging, device stability, environmental effects, and variations between individual devices of the same model should also be considered, but have not been assessed here. Device selection and purchasing advice are deliberately left out, as are cost, availability, market share, reliability, specification risk, batch consistency, time-keeping and actigraphy performance, battery life, lengths of logging period, and measurement intervals.

It will be frustrating to some not to have the answers to all of these questions, but it can be appreciated that what is provided is a useful advance, and perhaps the most important element, in that lengthy and challenging task. It is to be hoped that in the future CIE may provide for non-visual dosimeters what exists for illuminance and luminance meters.

2. METHODS AND PERFORMANCE METRICS

The spectral, directional, and dynamic responses were characterized using three separate measurement protocols. In each case, the responses to controlled light exposure conditions were recorded in temperature-controlled and darkened environments.

The 15 dosimeters, and the abbreviations adopted, are set out in Table 1. The methods described were carried out for seven of the models (MW8, ATR, GAO, AWL, AW2, AWS, AWSP). The data for the AWL are based on similar protocols described in a previous study [7]. The present study extends the range of dynamic response results of other protocols for the AWS in [7,8].

The LXB data for $V(\lambda)$ and $c(\lambda)$ sensors (referred to as V and c) in Hubalek *et al.* [6] are author data for underlying optical sensor components (S7686 and S6428-01, respectively, Hamamatsu, Hamamatsu City, JP) [14] and the bare components appear to be used in the finished device. The $V(\lambda)$ function for photopic vision is based on the spectral luminous efficacy of a monochromatic radiation at wavelength λ that is used to characterize photometers [4], i.e., it is the function that describes the relative visual sensitivity of the eye. Similarly, the Gall function $c(\lambda)$ is a non-standard empirical function based on wavelength sensitivity to melatonin suppression that has historically been used to represent non-visual sensitivity [6]. Its use is now deprecated [1,3].

The DIME incorporates another Hamamatsu optical sensor component, an IR-blocking filter and an opal diffuser. The unconfirmed spectral data for the assembled device were taken

Table 1. Make, Model, Abbreviation, Present Manufacturer, and Brief Scope of Presented Data for the Devices Identified[a]

Make, Model, and Abbreviation	Manufacturer	Data Included
ActiGraph wGT3X-BT (GT3X) and ActiSleep-BT (SLPB)	ActiGraph, Pensacola, US	Online data
CamNtech Motionwatch8 (MW8)	CamNtech, Cambridgeshire, UK	Complete protocol
ActTrust (ATR)	Condor Instruments, Sao Paolo, BR	Complete protocol
Daqtix Daqtometer (DAQ)	Daqtix GmbH, Oetzen, DE	No data to present
GeneActiv Original (GAO)	Activinsights Ltd, Cambridgeshire, UK	Complete protocol
LRC Daysimeter-D (DIME) and Daysimeter-S (DAYS)	Lighting Research Center (LRC), Troy, NY, US	Figueiro *et al.* [8] and online data
Actiwatch L (AWL)	No longer in production	Unpublished data, as Price *et al.* [7]
Actiwatch 2 (AW2)	Philips Healthcare, Best, NL	Complete protocol
Actiwatch Spectrum (AWS)	No longer in production	Price *et al.* [7], plus unpublished data
Actiwatch Spectrum Plus (AWSP) and Pro (AWS Pro)	Philips Healthcare, Best, NL	Complete protocol
LuxBlick (LXB)	See Hubalek *et al.* [6]	Hubalek *et al.* [6], plus online data
OT Lightwatcher (OTLW)	Object-Tracker, Perchtoldsdorf, AU	Online data

[a]Minimiter (US) and the Actiwatch brand was acquired by Philips Respironics, now itself a part of Philips Healthcare.

from its technical specifications [15], and the remaining data from Figueiro *et al.* [8]. For the GT3X and SLPB, spectral sensitivity data were taken from unconfirmed technical specifications of the component (TPS861, Toshiba, Tokyo, JP) [16] and exclude any effects of the input optics on spectral sensitivity in the finished devices. Online images suggest the sensor is positioned behind a transparent red case for the GT3X but not the SLPB, so there is a potential performance difference between these models. For the OTLW, the visible sensor spectral data were taken from unconfirmed data for the relevant components (S7686 as LXB V and S9032-02) [17]. The spectral data relates to the bare components. The PTFE aperture input optics should have no impact on the spectral sensitivity of the assembled device.

The ATR and OTLW infrared sensors and the OTLW ultraviolet sensor were excluded as this study concerns dosimetry for photoreceptive responses in the eye (see [17,18] for additional information).

A. Spectral Response

The devices were exposed to radiation from a purpose-built laboratory monochromator. Spectral bandwidth was balanced against throughput to minimize non-linearity. The resulting 26 light conditions covered a range of 360–800 nm with bandwidths between 12.5 and 15.5 nm full width at half-maximum. Exposures to additional wavelengths up to 930 nm were used to determine near-infrared sensitivity.

Ideally, the metrics applied to the six sensitivity curves should follow a consistent formula [19]. The CIE report's metric f_1', the "general $V(\lambda)$ mismatch index," involves area-normalization of the sensitivity curve against $V(\lambda)$ using CIE standard illuminant A as a reference spectrum, but neither $V(\lambda)$ nor illuminant A is generally relevant here. $V(\lambda)$ relates to vision rather than non-visual responses, and Illuminant A is relatively biased toward longer wavelengths; such tungsten-halogen spectra are also quite distinct from daylight and the increasing range of other artificial lighting technologies. f_1' is calculated as follows:

$$f_1' = 100\% \frac{\int_{380\,\mathrm{nm}}^{780\,\mathrm{nm}} |s_{\mathrm{rel}}^*(\lambda) - V(\lambda)|\mathrm{d}\lambda}{\int_{380\,\mathrm{nm}}^{780\,\mathrm{nm}} V(\lambda)\mathrm{d}\lambda}, \tag{1}$$

where $s_{\mathrm{rel}}^*(\lambda)$ is proportional to the sensor relative spectral sensitivity function $s_{\mathrm{rel}}(\lambda)$, and normalized to match $V(\lambda)$ according their responses to the calibration source spectrum $S_{\mathrm{A}}(\lambda)$, i.e.,

$$s_{\mathrm{rel}}^*(\lambda) = s_{\mathrm{rel}}(\lambda) \frac{\int_{380\,\mathrm{nm}}^{780\,\mathrm{nm}} S_{\mathrm{A}}(\lambda) V(\lambda)\mathrm{d}\lambda}{\int_{380\,\mathrm{nm}}^{780\,\mathrm{nm}} S_{\mathrm{A}}(\lambda) s_{\mathrm{rel}}(\lambda)\mathrm{d}\lambda}. \tag{2}$$

In order to generalize this method, the α-opic action spectra can be considered as conditional probability distributions $p(\lambda|E_{\lambda,\mathrm{E}})$, the condition being responses at each wavelength relate to a uniform spectral irradiance $E_{\lambda,\mathrm{E}}$. $p(\lambda|E_{\lambda,\mathrm{E}})$ denotes the probability density function for a photon (or classical differential irradiance), drawn from a uniform spectral irradiance, having the wavelength λ. The uniform irradiance $E_{\lambda,\mathrm{E}}$ can be substituted by any suitable reference function $E_{\lambda,\mathrm{R}}$. Potential substitute reference functions might be standard illuminant A, D65, or spectrally uniform photon flux per unit area. The use of $E_{\lambda,\mathrm{E}}$ has practical advantages mathematically (relied on later), physically (proportionality to its SI action spectrum) and subjectively (it is broadly considered "spectrally neutral" in various fields). As a result, they are consistently area-normalized against a spectrally uniform reference spectrum.

In general terms, percentage mismatch can be defined as the actual response over the expected or required response less one. This is only dimensionally achieved in the CIE metric f_1', so that mismatches of over 100% may arise and no meaningful formula for $f_1'(\lambda)$ is possible. Using probabilistic distributions in contrast, the spectral mismatch can be defined at any wavelength λ as

$${}_{\mathrm{p}}f_1(\lambda) = 100\%(s(\lambda|E_{\lambda,\mathrm{E}})/p(\lambda|E_{\lambda,\mathrm{E}}) - 1), \tag{3}$$

where $s(\lambda|E_{\lambda,\mathrm{E}})$ is the spectral sensitivity curve of the sensor with the area normalized to unity.

Spectral mismatch can be calculated using a simplified formula that produces 100% mismatch where the sensitivity curve and action spectrum do not overlap, and that produces zero for a perfect match:

$${}_{\mathrm{p}}f_1 = 100\% \int_0^\infty \frac{|s(\lambda|E_{\lambda,\mathrm{E}}) - p(\lambda|E_{\lambda,\mathrm{E}})|}{2}\mathrm{d}\lambda. \tag{4}$$

$|{}_{\mathrm{p}}f_1|$ has been weighted by the function $p(\lambda|E_{\lambda,\mathrm{E}})/2$, to emphasize mismatches near to the peak expected sensitivity and to normalize the resulting metric. Both ${}_{\mathrm{p}}f_1$ and f_1' rely on summations across a finite number of narrowband spectral measurement conditions identified by peak wavelength. ${}_{\mathrm{p}}f_1$ is also independent of the choice of calibration source. f_1' for $V(\lambda)$ is given for comparison purposes.

In summary, ${}_{\mathrm{p}}f_1$ is the spectral mismatch derived from f_1' using probabilistic distributions, substituting illuminant A by the uniform spectrum to act as a notional calibration source and finally dividing by 2 to limit the maximum mismatch to 100%. Using a notional calibration source allows for the integral limits to change to suit the application; unless specified below, the spectral range 380–780 nm is used.

Equation (4) provides a straightforward method to present spectral mismatches of broadband irradiance meters against different target action spectra. Note that like Eq. (1) it emphasizes departures around the wavelengths of peak sensitivity, but without the normalization bias toward longer wavelengths introduced by the use of illuminant A in Eq. (2).

The CIE standard [4] also defines a spectral mismatch correction factor (SMCF):

$$F^*(S_{\mathrm{Z}}(\lambda)) = \frac{\left(\frac{\int_{\lambda_{\min}}^{\lambda_{\max}} S_{\mathrm{Z}}(\lambda) s_{\mathrm{rel}}(\lambda)\mathrm{d}\lambda}{\int_{360\,\mathrm{nm}}^{830\,\mathrm{nm}} S_{\mathrm{Z}}(\lambda) V(\lambda)\mathrm{d}\lambda}\right)}{\left(\frac{\int_{\lambda_{\min}}^{\lambda_{\max}} S_{\mathrm{A}}(\lambda) s_{\mathrm{rel}}(\lambda)\mathrm{d}\lambda}{\int_{360\,\mathrm{nm}}^{830\,\mathrm{nm}} S_{\mathrm{A}}(\lambda) V(\lambda)\mathrm{d}\lambda}\right)}, \tag{5}$$

where $S_{\mathrm{Z}}(\lambda)$ is the relative spectral distribution of the light source being measured. $S_{\mathrm{Z}}(\lambda)$ can be substituted with the spectral irradiance E_λ of the light source without changing the result. Indeed, when the wavelength limits are matched the SMCF $F^*(S_{\mathrm{Z}}(\lambda)) = {}_{\mathrm{p}}F^*(E_\lambda)$, and it can be calculated as follows:

$$ {}_{\mathrm{p}}F^{*}(E_{\lambda}) = \frac{\int_0^{\infty} E_{\lambda}\, p(\lambda|E_{\lambda,\mathrm{E}})\mathrm{d}\lambda}{\int_0^{\infty} E_{\lambda}\, s(\lambda|E_{\lambda,\mathrm{E}})\mathrm{d}\lambda}. \tag{6} $$

Using either approach SMCFs quantify the mismatch correction for devices exposed to a general polychromatic light condition, whereas f_1' and ${}_{\mathrm{p}}f_1$ only quantify the mismatch for exposure to a uniform spectrum. If necessary, the $s(\lambda|E_{\lambda,\mathrm{E}})$ values provided in Dataset 1, Ref. [20] contain all the device-specific information required to calculate SMCF values.

Note that Eqs. (3), (4), and (6) apply equally to spectral radiance and other collection geometries if required.

The spectral responses of the devices have been assessed against each of the five α-opic non-visual action spectra $N_{\alpha}(\lambda)$ [3] (α = sc, z, r, mc, and lc for short wavelength cones, ipRGCs, rods, medium-wavelength cones, and long-wavelength cones, respectively, also see Table 2 and Fig. 1), and against the spectral luminous efficiency function $V(\lambda)$ for reference.

Prior to IWCNP, proposed circadian action spectra included an empirical function [21] based on nocturnal melatonin suppression studies [22,23], the melanopic sensitivity curve [24–26] governing the sensitivity of newly discovered intrinsically photosensitive retinal ganglion cells, and the theoretical composite circadian light function [27], later modified [28].

Although these models for human action spectra are deprecated by IWCNP, Price [11] subsequently suggested that many non-visual action spectra asymptotically approach the melanopic sensitivity curve with increasing exposure duration. Therefore, it was further suggested that melanopic sensitivity might be appropriate as the marginal action spectrum for circadian responses in naturalistic settings which imply light exposures of several hours. The melanopic function is one of the five α-opic functions, so the spectral analysis presented includes the results for following this simplified approach.

B. Directional Response

To determine the directional response, the devices were exposed to radiation from a green 532 nm laser (MGL-III-532, CNI Optoelectronics Tech. Co Ltd., PRC, 250 mW). The beam was expanded and collimated using a telescope with a ×40 micro-objective and a 85 mm focal length lens, to produce a uniform beam in excess of 12 mm in diameter. The beam was attenuated with neutral density filters and bracketing techniques were used when required to minimize non-linearity. An identical protocol using a blue 473 nm laser (MBL-III-473, 47 mW) was used in addition to the green laser protocol for selected devices.

The temporal variation of the laser beam was continuously monitored with a power meter (e.g., Coherent, Inc., Santa Clara, California, US) using a beam splitter. The instability of the laser beam entering the sphere was up to approximately 0.5%, so no corrections were applied.

Measurements were made at 2 deg intervals along the two halves of two semi-axes. The input optics were mounted at the axis of rotation, with the acceptance area, i.e., including the cross-sectional areas of both the input optics and the internal sensor components, overfilled at all angles.

The CIE directional response mismatch or error is given as

$$ f_2(\epsilon,\varphi) = 100\%\left(\frac{Y(\epsilon,\varphi)}{Y(\epsilon=0)\cos\epsilon} - 1\right) \tag{7} $$

and the metric for the departure from a perfect cosine response, f_2 "directional response error," as

$$ f_2 = 100\% \int_0^{\epsilon_{\max}} \left|\frac{Y(\epsilon,\varphi)}{Y(\epsilon=0)\cos\epsilon} - 1\right| \sin 2\epsilon\, \mathrm{d}\epsilon, \tag{8} $$

where ϵ is the angle to the optical axis, i.e., from the normal to the surface of the input optics, φ is the azimuth angle, and $Y(\epsilon,\varphi)$ is the response as a function of angle. The 1987 standard used 1.484 rad, i.e., 85 deg for the upper integration limit, and this approach is followed here for consistency with previous studies, otherwise 80 deg is to be used from 2014 [4]. As the performance metric f_2 is widely quoted for broadband cosine response sensors, it will be the main directional response metric quoted in Table 3 below.

Equation (8) is similar to Eq. (4), but incorporates a weighting factor proportional to sin 2ϵ that defines where the mismatch is considered most relevant. It is possible to derive a metric ${}_{\mathrm{p}}f_2$ using an analogous formula to ${}_{\mathrm{p}}f_1$ including this weighting, which is therefore limited to 100%. Although f_2 is not strictly limited to 100%, especially when the sensitivity increases with ϵ over part of the interval, it turns out that the new and old metrics are closely related, i.e., ${}_{\mathrm{p}}f_2 \approx f_2/2$.

The relation holds provided the response is approximately cosine, or when $\int_0^{\epsilon_{\max}} \frac{Y(\epsilon,\varphi)}{Y(\epsilon=0)\cos\epsilon}\sin 2\epsilon \mathrm{d}\epsilon \approx \int_0^{\epsilon_{\max}} \sin 2\epsilon \mathrm{d}\epsilon$. These two terms are the normalizing denominators needed on the respective two terms in the integrand in Eq. (8) to convert f_2 into twice ${}_{\mathrm{p}}f_2$.

A correction factor can be applied when measurements are carried out with different angular distributions of incident light than that desired, provided both angular distributions are known. As proposed here, the directional correction factor ${}_{\mathrm{p}}F_2^{*}$ is the ratio of summed relative directional responses over the summed cosine function:

$$ {}_{\mathrm{p}}F2^{*}(L_{\epsilon,\varphi}) = \frac{\int_0^{2\pi}\int_0^{\pi/2} L_{\epsilon,\varphi}\cos\epsilon\sin\epsilon\,\mathrm{d}\epsilon\mathrm{d}\varphi}{\int_0^{2\pi}\int_0^{\pi/2} L_{\epsilon,\varphi}\frac{Y(\epsilon,\varphi)}{Y(\epsilon=0)}\sin\epsilon\,\mathrm{d}\epsilon\mathrm{d}\varphi}, \tag{9} $$

where $\int_0^{2\pi}\int_0^{\pi/2}\sin\epsilon\,\mathrm{d}\epsilon\mathrm{d}\varphi = 2\pi$ steradians is the solid angle of the forward-facing hemisphere and $L_{\epsilon,\varphi}$ is the known angular distribution of source luminance. There are obvious similarities to Eq. (6), but also some differences to note, such as the double integrals in Eq. (9), as classical direction has two degrees of freedom, one more than wavelength.

C. Dynamic Response

The dynamic response quantifies the relationship between electrical output and light input. It is characteristic of photodiodes that their spectral sensitivity is additive and stable at any given temperature. With sufficient throughput, any spectrum can be used to determine the dynamic response and linear range of these sensors, including monochromatic sources. Spectral additivity also makes photodiodes suitable for broadband metrology, although other types of sensor may share this property.

1. Dynamic Resolution

Dynamic resolution is the smallest incremental difference in the response that can be recorded due to an increase in the quantity being measured. The term resolution is often used, but this could be ambiguous, especially if not qualified. The non-standard term dynamic resolution is considered preferable.

Dynamic resolution can usually be revealed using any large set of finely graduated data. The directional response data met this requirement. The data were sorted in ascending order. The non-zero differences between the sorted data were plotted against the corresponding sorted values on a log-log scale. The dynamic resolutions are given in Table 4 and in Fig. 3, where the dynamic resolutions are shown as a percentage of the sorted values.

All the devices in Table 1 are digital. Noise due to random variations arising from within the device, e.g., electronic noise, may be greater than the dynamic resolution. This procedure characterizes dynamic resolution for digital devices only when the discrete steps in possible responses are significantly larger than the noise. The directional response data showed whether a noiseless signal would result from a range of stable exposures. For the devices tested this indicated no further investigations were needed to quantify the contribution of noise to dynamic response errors, at least in temperature-controlled conditions.

2. Dynamic Range and Linearity

To determine the dynamic range and linearity, including the dark signal, the devices were exposed to radiation from the green laser described above. The input optics were aligned with the 10 mm diameter exit aperture of a small integrating sphere, similar to the method of Figueiro *et al.* [8,15] and filters were placed in the beam line prior to the integrating sphere and positioned to reduce the impact of stray light. The tested conditions covered over 8 orders of magnitude and twice the log-range of any previously published results for such devices.

The transmittance of the neutral density filters at 532 and 473 nm were ascertained using this sphere-based protocol.

This protocol is based on exposures to directionally homogeneous illumination, whereas Price *et al.* [7] measured the dynamic response of the devices when exposed to normally incident radiation with a uniform cross section, i.e., this represents the dynamic response to a different angular distribution of incident light. The relative advantages of using normally incident and directionally homogeneous illumination may depend on the purpose of the device; in many cases environmental exposures are directionally homogeneous. The CIE report specifies homogeneous illumination of the acceptance area, and elsewhere refers to this same source as vertical illumination, which strongly implies illumination which is not directionally homogeneous.

As $L_0 = L_{\epsilon,\varphi}$ is constant for directionally homogeneous light and $L_1 = L_{\epsilon,\varphi}$ is zero everywhere except for $L_{\epsilon=0,\varphi=0}$ for normally incident radiation, using Eq. (9) ${}_{\mathrm{p}}F_2^*(L_1) = 1$ and the correction factor for switching between two methods of calibrating dynamic responses can be calculated as ${}_{\mathrm{p}}F_2^*(L_0)/{}_{\mathrm{p}}F_2^*(L_1) = {}_{\mathrm{p}}F_2^*(L_{\epsilon,\varphi} = \text{constant})$. This is the value of ${}_{\mathrm{p}}F_2^*$ shown in Table 3. The integrals were realized as area summations up to $\epsilon = 1.484$ rad.

The analogous equation to Eqs. (3) and (7) for a dynamic mismatch factor is

$$ {}_{\mathrm{p}}f_3(Y) = 100\%\left(\frac{Y(X)}{Y(X_{\max})}\frac{X_{\max}}{X} - 1\right). \tag{10}$$

In a potential improvement to Eq. (10) the absolute value of ${}_{\mathrm{p}}f_3(Y)$ would be subject to a minimum of half the percentage dynamic resolution, i.e., $100\%\Delta Y(X)/2Y(X)$, as this reflects the accuracy to which X can be determined from Y.

It would be helpful to develop analogs of spectral Eqs. (4) and (6) for dynamic responses, just as Eqs. (8) and (9) deal with directional responses. Further photobiological data involving responses to variable levels of light are needed to ensure such metrics fairly reflect desired performance. The theoretical formulation of ${}_{\mathrm{p}}f_3$ currently under development was as low as 9% for one of the measured sensor combinations shown in Table 2 and most sensors returned below 20%. Spectral and potentially directional errors may be more important in this type of device. Investigating this is left for future work.

Rather than a monochromatic source, Price *et al.* [7] also used a broadband illumination to determine the dynamic response. Broadband sources can provide throughput for a wider range of spectral sensitivity functions using a single source, whereas when using narrowband sources or monochromatic laser sources, the throughput is improved by selecting a wavelength close to the peak sensitivity of a given sensor. Otherwise, provided that spectral additivity applies, the results should be interchangeable.

D. Important Note on the Performance Metrics

The limitations for mismatch correction factors ${}_{\mathrm{p}}F_2^*(E_\lambda)$ and ${}_{\mathrm{p}}F_2^*(L_{\epsilon,\varphi})$ should not be overlooked. These metrics apply only for known constant spectral irradiance, known constant angular luminance distributions, and known dose-weighted and α-opic weighted irradiance time series. They may be appropriate for specialized laboratory purposes, but cannot be applied to bulk field data, or only approximately.

${}_{\mathrm{p}}f_1$ and ${}_{\mathrm{p}}f_2$ (or f_2), however, can be used to assess dosimeters against performance targets. The proposal is for ${}_{\mathrm{p}}f_1$ and ${}_{\mathrm{p}}f_2$ to form the basis of a new standard for broadband dosimeters. Through the selection of appropriate weighting functions, the metrics are defined so that their values are directly comparable to the typical measurement errors, and so that the minimum value is 0% for the perfect performance and the worst possible performance provides the maximum value of 100%.

The metrics described have been used here for non-visual photometry, but can be applied more widely to optical dosimetry, e.g., to broadband meters used in photodynamic therapy [29].

3. RESULTS

A. Spectral Response

The spectral performance of the dosimeters are set out in Table 2, the action spectra $N_{\mathrm{sc}}(\lambda)$, $N_{\mathrm{z}}(\lambda)$, $N_{\mathrm{r}}(\lambda)$, $N_{\mathrm{mc}}(\lambda)$, $N_{\mathrm{lc}}(\lambda)$, and $V(\lambda)$ are shown in the top-left panel of Fig. 1, and the sensitivities of the sensors in the dosimeters measured are shown in the other panels.

Table 2. Percent Spectral Mismatches $_{\mathrm{p}}f_1$ to the Photopic $V(\lambda)$ and α-opic $N_\alpha(\lambda)$ Non-Visual Sensitivity Curves,[a] and f_1' to $V(\lambda)$

Model Sensor	Cyanopic $_{\mathrm{p}}f_1{:}N_{\mathrm{sc}}(\lambda)$	Melanopic $_{\mathrm{p}}f_1{:}N_z(\lambda)$	Rhodopic $_{\mathrm{p}}f_1{:}N_r(\lambda)$	Chloropic $_{\mathrm{p}}f_1{:}N_{\mathrm{mc}}(\lambda)$	Erythropic $_{\mathrm{p}}f_1{:}N_{\mathrm{lc}}(\lambda)$	Photopic $_{\mathrm{p}}f_1{:}V(\lambda)$	Photopic CIE $f_1'{:}V(\lambda)$
GT3X/SLPB	82	65	58	46	37	44	86
MW8	71	31	**15**	**17**	38	36	100
ATR				**12**	**17**		
ATR *Clear*	84	66	58	45	34	42	82
ATR Blue (B)	45	25	36	55	65	71	186
ATR Green (G)	90	60	44	**15**	**18**	**14**	27
ATR Red (R)	99	97	95	82	65	70	113
GAO	85	66	58	43	33	40	79
DIME			**15**	**5.9**	**11**		
DIME Min. f_1'	95	63	45	20	**16**	**12**	23
DIME Blue (B)	34	26	39	61	73	79	273
DIME Green (G)	88	51	32	**16**	30	25	64
DIME Red (R)	99	96	93	76	57	61	104
AWL	84	64	51	25	**17**	21	42
AW2	78	53	43	29	24	32	62
AWS		**16**	**12**				
AWS Blue (B)	31	39	49	67	77	83	315
AWS Green (G)	84	28	**18**	33	50	48	127
AWS Red (R)	99	98	96	87	72	76	122
AWSP[b]							
AWSP Blue (B)	135	112	98	70	51	51	75
AWSP Green (G)	71	74	74	69	56	64	121
AWSP Red (R)	127	125	121	104	84	89	124
LXB				**15**	**7.9**	**3.3**	
LXB Circadian (c)	29	31	45	66	78	83	303
LXB Visual (V)	95	68	51	23	**12**	**3.7**	7.3
OTLW			**17**	**16**	**19**	**16**	
OTLW Blue (B)	29	31	45	66	78	83	303
OTLW Green (G)	94	61	43	28	37	28	68
OTLW Red (R)	99	98	97	87	70	75	119

[a]Bold figures show $_{\mathrm{p}}f_1$ spectral mismatches below 20%.
[b]Based on marginal spectral sensitivity, see main text for further details.

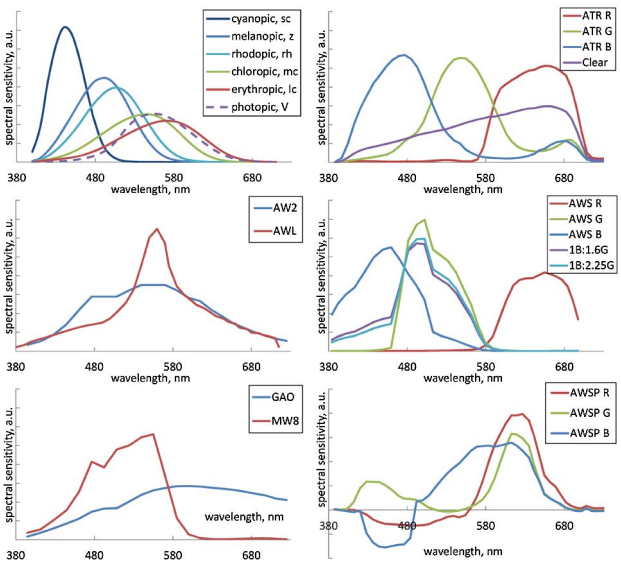


Fig. 1. Spectral sensitivity in arbitrary units (a.u.) plotted against wavelength in nanometers. Within each panel, areas below the lines are normalized for ease of comparison. The visual photopic and the five non-visual sensitivity functions are shown (top-left). The AWS (mid-right panel) includes linear sensor combinations optimized to match the melanopic and rhodopic curves, respectively.

Spectral mismatches would be expected between the sensitivity of devices based on previously suggested sensitivity curves and the current recommendations, unless the old and new target curves match (e.g., the old melanopic function). Where possible, previous data in the literature have been adapted to make assessments of spectral mismatches.

The spectral mismatch between the rhodopic and melanopic sensitivity functions is 16% and between erythropic and chloropic the mismatch is higher at 21%. To distinguish between the effects of the different photoreceptors of the eye, either spectral data is required, which is currently impractical in field studies, or smaller spectral mismatches must be achieved in dosimeters which use broadband sensors. For this reason a spectral mismatch of 20% has been used as a performance threshold.

A stricter approach would be to seek a better match than provided by the average of the two adjacent sensitivity curves (e.g., rhodopic and melanopic), which would imply a performance threshold of approximately 8%. Replacing $p(\lambda|E_{\lambda,\mathrm{e}})$ with the average of $p(\lambda|E_{\lambda,\mathrm{e}})$ and $s(\lambda|E_{\lambda,\mathrm{e}})$ in the integral of Eq. (4) exactly halves the value of $_{\mathrm{p}}f_1$. The average of $p(\lambda|E_{\lambda,\mathrm{e}})$ and $s(\lambda|E_{\lambda,\mathrm{e}})$ is the curve with the lowest common mismatch to both $p(\lambda|E_{\lambda,\mathrm{e}})$ and $s(\lambda|E_{\lambda,\mathrm{e}})$. This is the basis of several important mathematical and interpretative advantages of $_{\mathrm{p}}f_1$ compared to f_1', due to combining probability

normalizations with the equi-energy reference spectrum $E_{\lambda,e}$. The α-opic mismatches for adjacent spectral sensitivities are ${}_pf_1(\mathrm{sc, z}) = 58\%$, ${}_pf_1(\mathrm{z, r}) = 16\%$, ${}_pf_1(\mathrm{r, mc}) = 32\%$, ${}_pf_1(\mathrm{mc, lc}) = 21\%$, and additionally ${}_pf_1(\mathrm{photopic, mc}) = 20\%$ and ${}_pf_1(\mathrm{photopic, lc}) = 11\%$. This is arguably the performance threshold to support the meta-studies envisaged by IWCNP, but is only achieved for one, or at best two, of the sensitivity curves by any device. Before adopting this stricter approach, it might prove useful to test the impact of varying the assumptions underlying the α-opic sensitivity curves, considering in particular changes in the lens transmittance with age and inter-individual variability; this goes beyond the scope here.

Table 2 shows that the cyanopic curve is not well matched by any device, the closest being the LXB *c* and OTLW B sensors. These two sensors share an optical component, and the results were identical, except the OTLW B cyanopic mismatch was calculated using data extracted from a graph to be 30%, so the result using the tabulated LXB data, 29%, has been quoted.

Of the devices measured in this study, the best match to spectral sensitivities of melanopic and rhodopic curves are from the AWS based on positive linear combinations of the AWS B and G sensors in ratios of 1:1.6 and 1:2.25, respectively. The former result updates the ratio to match the melanopic curve used in [7] as the new melanopic function [1] has a slightly longer peak wavelength (490 nm cf. 476 nm). These two combined sensitivities are not greatly independent (with a mutual mismatch of 4.4% cf. 16% between the melanopic and rhodopic curves). The AWS was the only device to provide a mismatch below 20% for the melanopic curve. The ATR was confirmed to have the closest match to the chloropic and erythropic curves using linear sensor combinations. Note that the third AWL and ATR devices had different sensitivities to the other AWL and ATR devices, respectively. The α-opic mismatches were 10% to 23% worse for the third AWL and 20% worse to 16% better for the combined sensor responses of the third ATR. They were excluded in Table 2 and Fig. 1.

It can be seen from Table 2 that, respectively, the MW8, AWL, AW2, and GAO spectral sensitivities were increasingly mismatched with the target sensitivity curves.

When combining data from more than one sensor, it is important to consider their independence spectrally and their linearity. At least two sensors with non-zero contributions are required to match any two curves independently, three is the minimum to match any three curves, and so on. The exact mathematical requirement would be a non-zero determinant of a coefficient matrix. Linear combinations of individual sensors will combine their dynamic responses. This may smooth out some non-linearities, but is likely to reduce the dynamic range over which the combined response is linear, when compared to the individual sensors.

The best acceptable mismatches from single sensors were the MW8 for the rhodopic curve and the ATR for the chloropic and erythropic curves. The AWL also provided an acceptable single sensor mismatch to the erythropic curve.

When including the models not measured here, the published DIME data [8,15] had the closest match to the chloropic curve and the mismatches to the erythropic and rhodopic curves were 11% and 15% using optimized linear combinations of sensors, which could be exploited providing the raw data are accessible. Note that the DIME Min. f_1' data is the built-in combined sensor response reported in [8] for matching $V(\lambda)$. The LXB [6] had the lowest mismatch to the erythropic curve, which was 7.9% and produced the closest match to any curve, with a mismatch of 3.3% to the photopic curve. In fact, the single V sensor mismatch of 3.7% in the LXB outperformed linear combinations from any other device measured. The DIME and LXB have acceptable single sensor mismatches to the chloropic and erythropic curves, respectively.

None of the devices had a single sensor mismatch to the melanopic curve below 20%. Currently, researchers interested in matching this important curve may have to combine sensors from the AWS.

On average, f_1' is around twice ${}_pf_1$ but they are not strictly proportional due to the spectral bias in f_1'. This metric overemphasizes any mismatches from the target sensitivity $V(\lambda)$ at longer wavelengths, by more than 0.7% per nm compounding at 555 nm. ${}_pf_1$ is, arguably, easier to interpret and is spectrally neutral. There would be problems generalizing f_1' to a target sensitivity with a peak wavelength below around 500 nm, as the bias increases when applied to curves with lower peak wavelengths. Figure 1 shows why the MW8 scores poorly on this statistic against $V(\lambda)$, as the sensitivity above 580 nm is almost zero, similar to the green sensor on the AWS. Subjectively, it is not obvious that the MW8 has a worse match to $V(\lambda)$ than the GAO (see Fig. 1), as suggested by their measured f_1' values.

1. Marginal Spectral Sensitivity for the AWSP

A different approach was needed for the AWSP, so the mismatches shown in Table 2 were based on marginal sensitivity curves. Under narrowband light, the AWSP RGB spectral sensitivities abruptly dropped to zero for several wavelengths. This was thought to be due to pre-programmed interactions between the three photodiodes and not observed in other dosimeters. Whatever the cause, it was suspected that the spectral responses were not additive; spectral additivity is an important property of the individual α-opic curves, as stated in Section 2.C.

To test this, a stable background broadband light (tungsten-halogen GU10 spot, retail product) was added to the exposures of the monochromator using a beam splitter. Wavelengths above 668 nm were filtered out and this background source provided between 2.6 to 10 times the spectral irradiance of the monochromator (272 lx, CCT 3127 K, due to IR filter). The marginal spectral sensitivity was calculated in these conditions.

For some wavelengths the marginal spectral sensitivity was negative, as the response reduced with the addition of certain wavelengths from the monochromator, as shown in the bottom-right panel of Fig. 1. This confirmed the non-additivity of spectral responses for the AWSP.

However, the spectral mismatch statistics can still be calculated provided the integrated sensitivity curves are positive overall, and these data are quoted in Table 2. The presence of negative sensitivity values permits ${}_pf_1$ values to exceed 100% in some cases. It can also be seen from Fig. 1 that the "Red," "Green," "Blue" labels do not readily describe the spectral sensitivity of these sensors. In all cases, the spectral matches of the RGB sensors to the α-opic curves were approximately equal to or worse than a theoretical sensor with spectrally uniform

sensitivity. The AWSP spectral sensitivity was based on the performance of a single loan dosimeter, so it is not possible to be sure whether this was representative of the AWSP and AWS Pro models.

Had the mismatches been lower, the use of these sensors to match any of the α-opic functions would remain highly limited due to the non-additivity. These results cannot necessarily be interpreted as the marginal spectral sensitivity for other background conditions, e.g., daylight, fluorescent, or LED lighting.

2. Sensitivity to Out-of-Band Radiation

The ATR B and G sensors have out-of-range sensitivity, shown in the top-right panel of Fig. 1. The secondary peak sensitivities at approximately 680 nm compromise the matches to the melanopic and rhodopic sensitivity curves.

Suitable filters were used to test for unwanted sensitivity to infrared radiation using the monochromator for wavelengths up to 930 nm. The GAO (absolute) spectral sensitivity increases and remains significant approaching and into the IR. The AWSP R sensor exhibited spectral sensitivity to IR. Manufacturer data for the ATR [18] show marginal sensitivity in the IR region up to 1100 nm for all three RGB and Clear sensors, which was confirmed up to 930 nm by direct measurements.

The spectral data shown in Fig. 1 were considered sufficient to establish that the possibility of ultraviolet sensitivity was negligible for all the dosimeters measured.

B. Directional Response

Table 3 sets out the metrics for directional performance and the directional responses are shown in Fig. 2.

Table 3. Average Cosine Response Error f_2 ($_pf_2$) at 532 nm for Each Model, Plus Semi-Axes T2B, from Top to Bottom, and R2L, from Right to Left, and the Impact of the Dynamic Calibration Geometry, $_pF_2^{*a}$

	f_2 ($_pf_2$)			$_pF_2^*$
Model Sensor	Combined Axes	T2B Axes	R2L Axes	Combined Axes
MW8	46 (32)	46	46	174
ATR B	32 **(16)**	34	30	140
ATR G	29 **(15)**	29	29	**119**
ATR R	33 **(16)**	36	29	141
GAO	24 **(10)**	27	22	**91**
AWL	25 **(14)**	23	26	132
AW2	40 **(18)**	40	39	**94**
DIME B	**10** [8] (–)	–	–	–
DIME G	**10** [8] (–)	–	–	–
DIME R	**12** [8] (–)	–	–	–
AWS B	57 [7] (41)	49, 57 [7,8]	65 [7]	196
AWS G	47 [7] (32)	33, 39 [7,8]	61 [7]	160
AWS R	47 [7] (32)	33, 31 [7,8]	61 [7]	163
AWSP B	80 (70)	80	81	460
AWSP G[b]	80 (70)	80	79	467
AWSP R	79 (70)	79	79	536
LXB: V, 4870 K	**19** [14] (–)	–	–	–
c, 4870 K	**17** [14] (–)	–	–	–
c, 2856 K	114 [14] (–)	–	–	–

[a]Bold figures show combined $_pf_2$ errors within 20%, acceptable f_2 values if $_pf_2$ is not available, and $_pF_2^*$ between 80% and 120%.

[b]Excludes outlier for light at 28 deg below the normal, see Fig. 2.

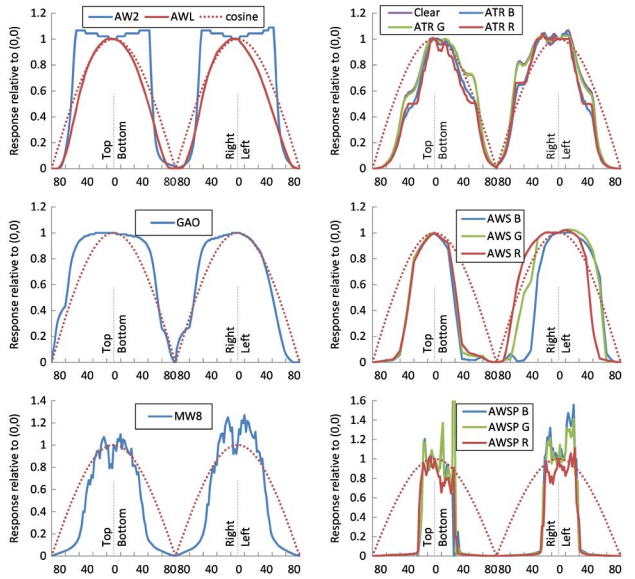


Fig. 2. Directional responses of the dosimeters tested compared to an ideal cosine response (selected angles in degrees, incident from the watch face top, bottom, right, and left in plan view, are labeled on the horizontal axes). Note the change in vertical scale for MW8 and AWSP for presentational purposes only. AWL and AWS curves are based on linear interpolation of $f_2(\epsilon, \varphi)$ of responses at 10 deg intervals.

The f_2 values in Table 3 do not take account of a minor change in ISO/CIE 19476:2014 to limit the assessed angles to 80 deg rather than 85 deg. In order to make comparisons to previous figures, the 85 deg limit has been retained, and used to calculate $_pf_2$ values, which are shown in brackets. The impact of this change depends on the departure from the ideal cosine response between 80 and 85 deg for each device.

Percentage cosine response errors were almost in single figures for the DIME. From the commercially available devices, the AWL, GAO, and ATR G errors were within 30%, but for the GAO this value is somewhat distorted by non-linearities (see later). As before, the chosen threshold of a 20% error for $_pf_2$ is certainly a large mismatch to permit; in the CIE report in 1983 the representative f_2 value for the best commercial illuminance meters was 1.5%, but it distinguishes between the better and worse performing devices. In most cases, the older designs performed better.

Using the blue laser, the directional response of the GAO was observed to depend on wavelength, and this may be equally true for other devices. Similarly, the cosine response of the LXB c sensor for broadband sources depends on color temperature [6,14]. Together with mitigating parallax issues in other devices, as noted in Price *et al.* [7], this reinforces the advantages of using diffusers with photodiodes.

Of the devices measured, AWL, GAO, and MW8 have diffusers but quite different directional responses are apparent. The GAO and MW8 diffusers were relatively clear on visual inspection. As well as lacking diffusers, the AWS, AWSP, and AW2 responses are dominated by parallax, which also features noticeably in the MW8 response.

The difference in AWS angular responses between papers [7] ($n = 16$) and [8] is possibly due to different positioning of the recessed sensors relative to the window between dosimeters and other variations in individual devices.

C. Dynamic Resolution, Range, and Linearity

Figure 3 illustrates dynamic resolution in native device units. Dynamic resolution is 100% at the lowest non-zero value a device reports, and typically falls as the inverse of the output.

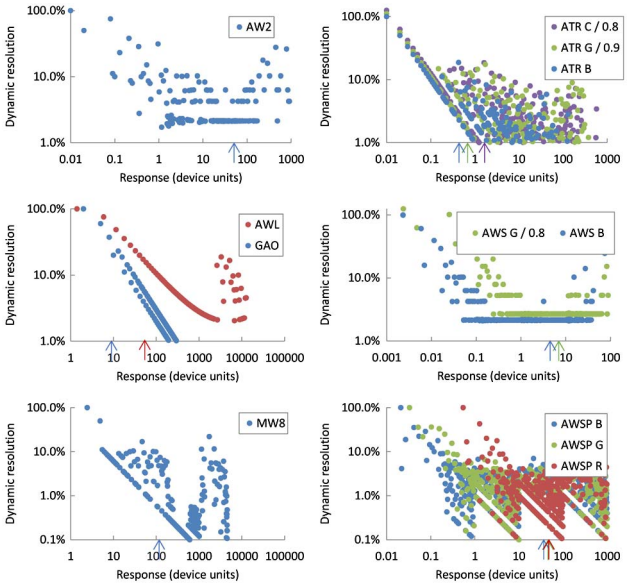


Fig. 3. Sorted difference data, illustrating dynamic resolution of the sensors, defined by the lowest achievable percentage for a given response. Vertical scaling factors and changes in both scales are for presentational purposes. Arrows show theoretical sensor responses to 100 lx of equi-energy light (0.137 $W \cdot m^{-2} \cdot nm^{-1}$). Note that the arrows appear at different values as the relationship between device units and illuminance differs for each sensor.

This is consistent with storing data to a fixed number of decimal places. Eventually, the dynamic resolution stops falling and reaches a limit or limiting pattern, either due to the number of significant figures stored by the device or truncating data beyond a target level of accuracy, i.e., for data compression.

The AWL is a clear exception, suggesting the values stored are linearly corrected only when they are downloaded. The MW8 and ATR had no clear limiting patterns, although there was evidence of data compression at relatively high values. No limiting pattern was observed in the GAO data, although the dynamic response was increasingly saturated at high values, effectively limiting the size of recorded values relative to the dynamic resolution to approximately 1% over 2 orders of magnitude of irradiances.

The lowest non-zero value should be sufficiently below the quantity or quantities to be measured. It is recommended the dynamic resolution be at least 1 order of magnitude below the total device units reported for the smallest quantity to be measured, to ensure an error contribution no greater than 10%. Dynamic resolution appears to be an avoidable source of measurement error, so 20% was not considered appropriate.

The arrows in Fig. 3 indicate the reported device units for a mid-range exposure. It is clear that the measured GAO and AWL devices did not satisfy the above requirement. The MW8, ATR B, and AWSP R only satisfy this requirement down to of 21, 3, and 12 lx of broadband light, respectively (or the same values of equivalent illuminance, also in lx, see below). All the other sensors tested here had an adequate dynamic resolution for the proposed target linear range.

The AW2, AWS, and, possibly, the AWSP appear to have a banded descent down to the lower threshold, i.e., a region where the minimum percent dynamic resolution is variable. An estimate of the worst dynamic resolution from the band is given for these devices in this region, and this was usually found to be sufficiently low.

Table 4 sets out the dynamic resolution and linear ranges. Unweighted irradiance is reported to enable test conditions to

Table 4. Dynamic Performances of the Devices, Shown as Equivalent Illuminances and Unweighted Irradiances, Relating to Test Conditions from 1.77 μW · m⁻² to 284 W · m⁻²

Model Sensor	Linear Range,[a] Equivalent Illuminance, lx	Linear Range Irradiance, W · m⁻²	Dynamic Resolution Equivalent Illuminance, lx	Dynamic Resolution Device Units
MW8	3.82 to 98,340	0.00552 to 183	(2.1, 8.2) 16,000 *i.e., 2.1 below 16,000 and 8.2 above 16,000*	(0.0014, 0.0055) 11.2
ATR B	**1.15** to ≥47, 493	**0.00552** to ≥284	(0.23, **0.69**) 3161	(0.01, 0.03) 137
ATR G	**1.15** to ≥136, 269	**0.00182** to ≥284	**(0.14, 0.43)** 1962	**(0.01, 0.03)** 137
ATR R	**0.11** to ≥4693	**0.00552** to ≥284	**(0.17, 0.50)** 2291	(0.01, 0.03) 137
GAO	294 to 3099	0.900 to 11.3	(>10%, <**10**%) 339 ~1% at high values	(>10%, <**10**%) 302 or 3 units
DIME G	≤**1.5** to ≥21, 316	≤**0.002** to ≥31	No data available	No data available
AWL	No data available	No data available	(>10%, <**10**%) 2052% min. at ~3756	(>10%, <**10**%) 109 2% min. at ~2000
AW2	**1.62** to ≥106, 274	**0.00552** to ≥284	**(0.02, 2.2%)** 1.1	**(0.01, 2.2%)** 0.55
AWS B	**0.14** to ≥48, 261	**0.00182** to ≥361	**(0.13, 2.2%)** 3.0	**(0.0066, 2.2%)** 0.15
AWS G	**1.44** to ≫**200k**	**0.00182** to ≫**229**	(0.53, **2.2%**) 6.5	(0.041, **2.2%**) 0.50
AWS R	2.05 to ≥1921	0.272 to ≥284	**(0.11, 2.2%)** 2.0	**(0.011, 2.2%)** 0.20
AWSP G	**0.12** to ≥20, 960	**0.00182** to ≥284	(**0.023**, <**1**%)2.3	(**0.01**, <**1**%)1.0

[a]Over the linear range $|_p f_3(Y)| < 20\%$ relative to an appropriate reference value. Bold values indicate linear ranges satisfying the upper or lower σ-opic thresholds. No data were available for DIME *B* and *R* sensors.

be easily reproduced. However, using this or any other common quantity for different sensors presents a problem; when expressed in SI units, the desired dynamic thresholds are different for each sensor, depending on the action spectrum being matched.

A simple scheme to cope with this apparent complexity is to express the conditions used for testing dynamic response and linearity thresholds as equivalent illuminance. The equivalent illuminance (lx) for a sensor denoted by σ is derived from its probability-normalized sensitivity curve $s_\sigma(\lambda|E_{\lambda,\mathrm{E}})$ as follows:

$$E_\sigma = K_\mathrm{N} \int_0^\infty E_\lambda s_\sigma(\lambda|E_{\lambda,\mathrm{E}})\mathrm{d}\lambda, \tag{11}$$

where constant $K_\mathrm{N} \approx 73,000\ \mathrm{lm \cdot W^{-1}}$ and E_λ is the incident spectral irradiance. The subscript σ is used to denote a generalized sensor just as α is reserved to denote one of five human photoreceptor types.

Equivalent illuminances are readily comparable, whereas weighted-irradiance values are difficult to compare as they depend on the relative bandwidth and the shape of the sensitivity curves.

A linear range up to an equivalent illuminance of 200,000 lx includes the majority of typical exposure conditions and darkened conditions of greater than an equivalent illuminance of 2 lx at the eye between phase-shifting or melatonin-suppressing light stimuli are typically used in laboratory tests as sufficiently low to rule out significant photobiological effects [30]. Studies of light levels during sleep and at night may require lower thresholds.

A fixed dynamic resolution should be less than 10% of the lowest reported value in the dynamic range it applies to, or alternatively correspond to no more than an equivalent 0.2 lx throughout the entire range.

The blue laser at 473 nm was used in an attempt to increase some of the tested ranges for the AWS B and ATR B sensors. Over a fourfold increase would be needed to reach an equivalent melanopic illuminance of 200,000 lx. This approach extended the response range for the AWS B by only around 27% (to 8.3 orders of magnitude), but due to the lower power to sensitivity ratio of the available blue laser to the ATR B sensor, it was not sufficient to extend this sensor's results.

No other spectrally additive sensors measured had higher sensitivity at 473 nm than 532 nm. 532 nm would be most inefficient for cyanopic sensitivity, but none of the sensors tested were well matched to cyanopic. A slightly longer wavelength laser with similar power could be used to increase throughput for testing photopic, erythropic, and chloropic sensitivity.

Following further investigations, all the sensors except GAO and MW8 may be found to satisfy the higher threshold. Similarly, the AWS R sensor is likely to satisfy the lower threshold. Many of the sensors may meet the dynamic response requirements in full, noting that AWS G very nearly does on the basis of the data here.

GAO responses had the smallest linear range, but the dynamic response appeared to be a smooth increasing function over a wider range. It may be possible to generate linearity corrections, but this is left for future work. The MW8 and AWS sensors each had a step in its dynamic response between the conditions 28.4 $\mathrm{W \cdot m^{-2}}$ and 47.5 $\mathrm{W \cdot m^{-2}}$.

Figure 4 illustrates the linearity of the dynamic responses and the ranges tested in unweighted irradiance. ATR G and

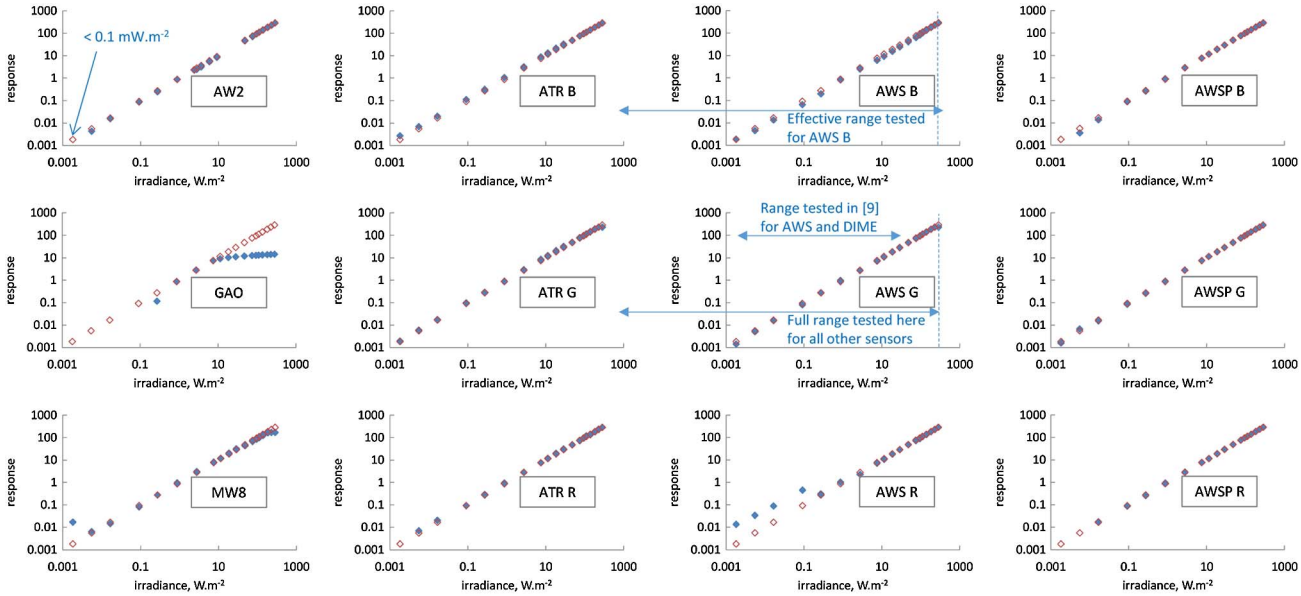


Fig. 4. Dynamic responses of the devices to 532 nm directionally homogenous radiation (solid blue diamonds) versus ideal linearity (red outline diamonds). The responses are scaled to agree the irradiance based on the maximum or an appropriate reference point within the linear range. From top to bottom: far-left AW2, GAO, MW8; mid-left ATR B, G, R; mid-right AWS B, G, R; far-right AWSP B, G, R. No AWL dynamic response data available. Only zero responses are not shown. Irradiance test conditions ranged from 1.77 $\mu\mathrm{W \cdot m^{-2}}$ to 284 $\mathrm{W \cdot m^{-2}}$, but lower limit shown is 1 $\mathrm{mW \cdot m^{-2}}$ as no dynamic ranges extended below this point. Some devices may be linear and/or unsaturated above the tested conditions. For all sensors $r^2 \geq 0.999$ except GAO with $r^2 \geq 0.760$ and MW8 with $r^2 \geq 0.979$ mainly due to saturation. Selecting an appropriate reference point for the GAO gives $r^2 \geq 0.999$ for responses up to 7.5 $\mathrm{W \cdot m^{-2}}$ and for the MW8 gives $r^2 \geq 0.999$ for responses up to 138 $\mathrm{W \cdot m^{-2}}$.

AWS G sensors exhibited more minor saturation effects than the MW8 and GAO. These are hardly noticeable on log-log linearity graphs, such as those in Fig. 4, nor in correlation coefficients calculated over many orders of magnitude.

Occasionally, one of the AWS G sensors was saturated. No output was given for any other sensor in the same epoch, including for conditions within the operating range of the other sensors. In fact, this has been observed in field data using AWS devices. Saturation effects may sometimes be corrected, e.g., if a smooth increasing dynamic response remains, or if the photobiological effect being considered is also insensitive to increases in light levels beyond a certain point.

4. OTHER DEVICE MODELS

A third-party loan of three DAQs was received, but the dosimeters could not be operated, due to technical difficulties. The DAQ responses may be modifiable by the user, with filters, diffusers, or both as the watch face is easily removable.

The objective was to identify suitable dosimeters for use in unobtrusive naturalistic personal dosimetry studies. The LXB [6,14], DAYS [8,15], and OTLW [17] were not included as the sensors are placed at the end of an elongated form, and/or were only consistent with mounting next to the eyes, rather than on a flat surface for wearing on the wrist or as a chest badge. It should be noted that there are, nevertheless, obvious advantages of both types of design. Note that in [8] the stated performance of the DAYS matches the DIME. It was decided, therefore, to report the percentage spectral mismatches based on the published technical information for these devices.

According to the manufacturer, reported illuminance from the GT3X and SLPB is capped at levels between 2500 and 6500 lx [16], and this is described as being unsuitable for outdoor measurements. Rather than exclude these models, it was decided to report the percentage spectral mismatches based on the published technical information for the sensor component. The significance of the difference in the input optics between the GT3X and SLPB is described in Section 2.

There are also a number of other related models that may have identical or similar performance. The GAO tested here is the GeneActiv Original or Model 1.1. Current GeneActiv models all have identical specification, sensors, and input optics. Any differences in input optics between the DIME and DAYS could create performance differences. However, according to [8], they have identical spectral, directional, and dynamic performance. The AWSP and the AWS Pro have identical specifications, so only the AWSP was measured. Contrary to assurances, however, the AWSP input optics and sensor performance bore very little relation to the original AWS, and cannot be considered a direct substitute.

Many other broadband light-logging devices exist, which have been excluded from our study on the basis of lacking validated research-grade actigraphy capabilities.

5. CONCLUSIONS

Optical performance was found to be quite varied, with each device having advantages and disadvantages. Whether a device is adequate will depend on the accuracy and the sensitivity curve(s) required by the user. The devices investigated do not currently support the meta-studies recommended by IWCNP.

New standards for these instruments have the potential to lead to greatly improved metrology in circadian dosimetry studies, if they were to become widely accepted and adopted. They could also help researchers understand the link between the design of dosimeters and the reliability of the data used to test light exposure theories in non-visual physiology. The metrics need not be limited to non-visual photometry. For consistency in photobiology, any new standards and metrics would ideally apply more widely to optical dosimetry, e.g., to broadband meters used in photodynamic therapy [29].

Where possible, more than one device should be tested to exclude unrepresentative performance. However, any sample or batch tested may not be representative of past or future dosimeters of the same model. This situation would also be different if the specifications were supported by appropriate standards for assessing the performance of non-visual dosimeters.

It is considered that a single device with multiple sensors matching the non-visual sensitivities of the human eye [1] would be the most useful upgrade that could be introduced in dosimeters for studies into links between sleep and circadian rhythms with ambient light. Provided the spectral sensitivities are appropriate, a lower number of multiple light sensors are likely to be advantageous over single-light sensors. Diffusers or beam-shaping optics may help improve angular responses.

Reliable validation of the actigraphy measurements is also necessary. Although this can be done by comparing analyses by proprietary software, e.g., [31], access to raw measurement data rather than the output of unpublished algorithms would enable independent validation. Built-in algorithms may discard useful data so that the opportunity for improving the analysis is lost. Even if optional, the algorithms in proprietary software amount to theoretical models of what patterns in these data may be most biologically relevant, so it would be appropriate for these to be published by the biological research community, rather than for the models to remain as commercially sensitive information. This is precisely the approach for the analysis of light time series a number of studies have already taken, notably [11–13].

When selecting a device to measure light exposure in a particular study, there are a number of other factors that have not been considered here, including cost, availability, reliability, specification risk, time-keeping, actigraphy performance, battery life, lengths of logging period, and measurement intervals. The literature covers batch consistency of the optical measurements only for the AWS [7,9], and this may be one of the most important considerations. Dataset 1, Ref. [20] can also be consulted to see the variability between devices, noting that for some models only one or two devices were measured.

The most likely source of batch inconsistency is probably physical device assembly rather than the underlying component quality. Variations in sensor position will often affect directional response directly, and linearity indirectly. This may be partly mitigated by a design with a good diffuser. Another variation may arise where component specifications change. In either case a researcher should seek a detailed specification wherever possible. The metrics introduced here are intended to give a useful structure for such detailed specifications.

Aside from the other factors, the researcher might start by eliminating devices with an insufficient linear range, also taking the required dynamic resolution into account when determining the lower linear range. The next step would be to consider what trade-off between directional and spectral mismatches would be appropriate. This is assisted by the fact that the new metrics ${}_{\mathrm{p}}f_1$ and ${}_{\mathrm{p}}f_2$ are directly comparable. The trade-off will also depend on how the measurements are aggregated in the study analysis. For example, if the light conditions will be averaged over an hour, and if the directional distribution of the light often changes considerably over this time frame, this will help to mitigate a large directional mismatch [8], so that it may be more important to select a low spectral mismatch device.

The importance of measurement is too often overlooked in epidemiological studies, where time outdoors may be taken as a proxy variable for UVR exposure in one study and for a psychological agent in another, with neither considering the many confounding exposures associated with the actual variable. The effect of time outdoors on light exposures and circadian rhythms is not trivial and may be the dominating influence on many outcomes in studies of health and well-being.

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